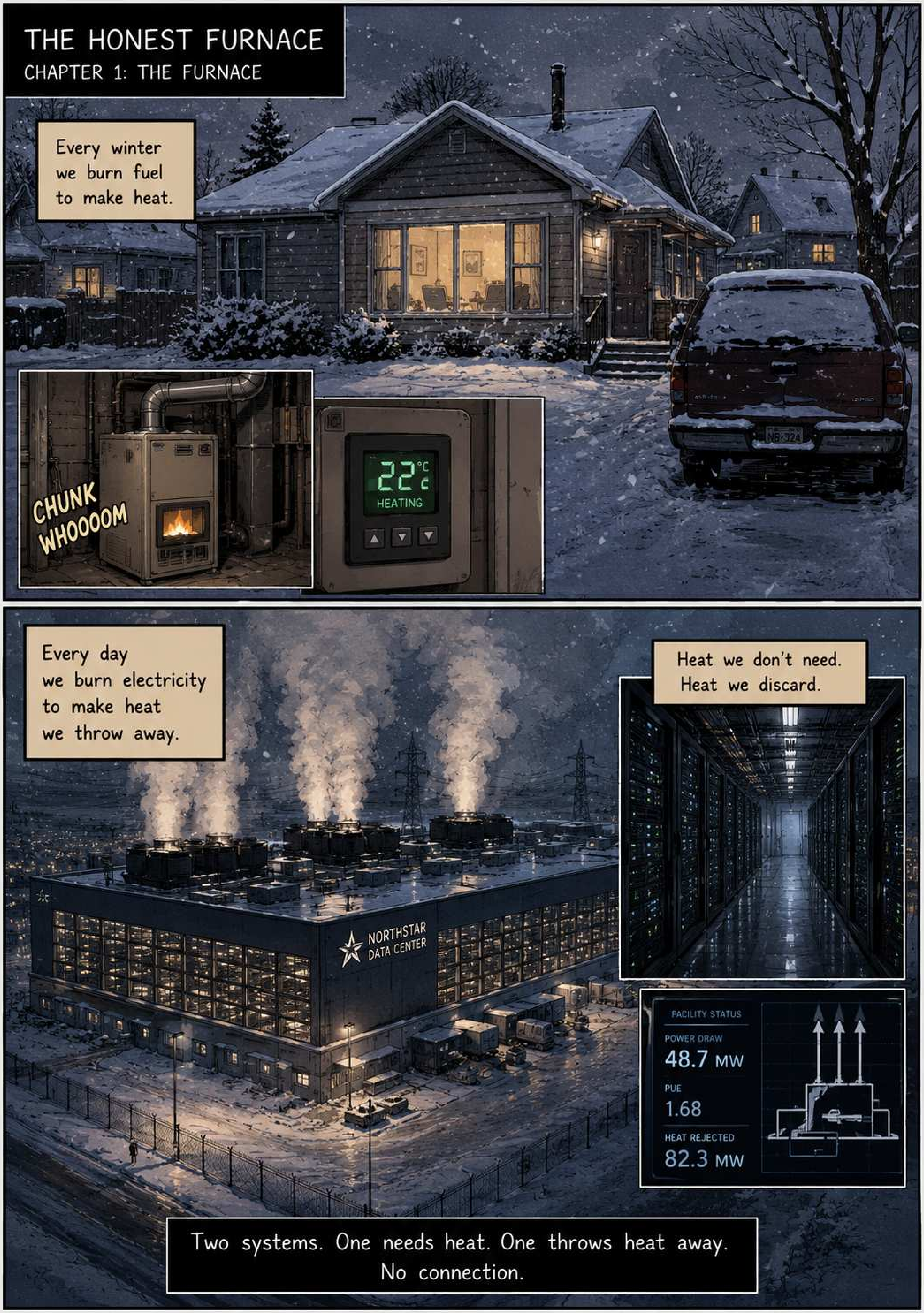
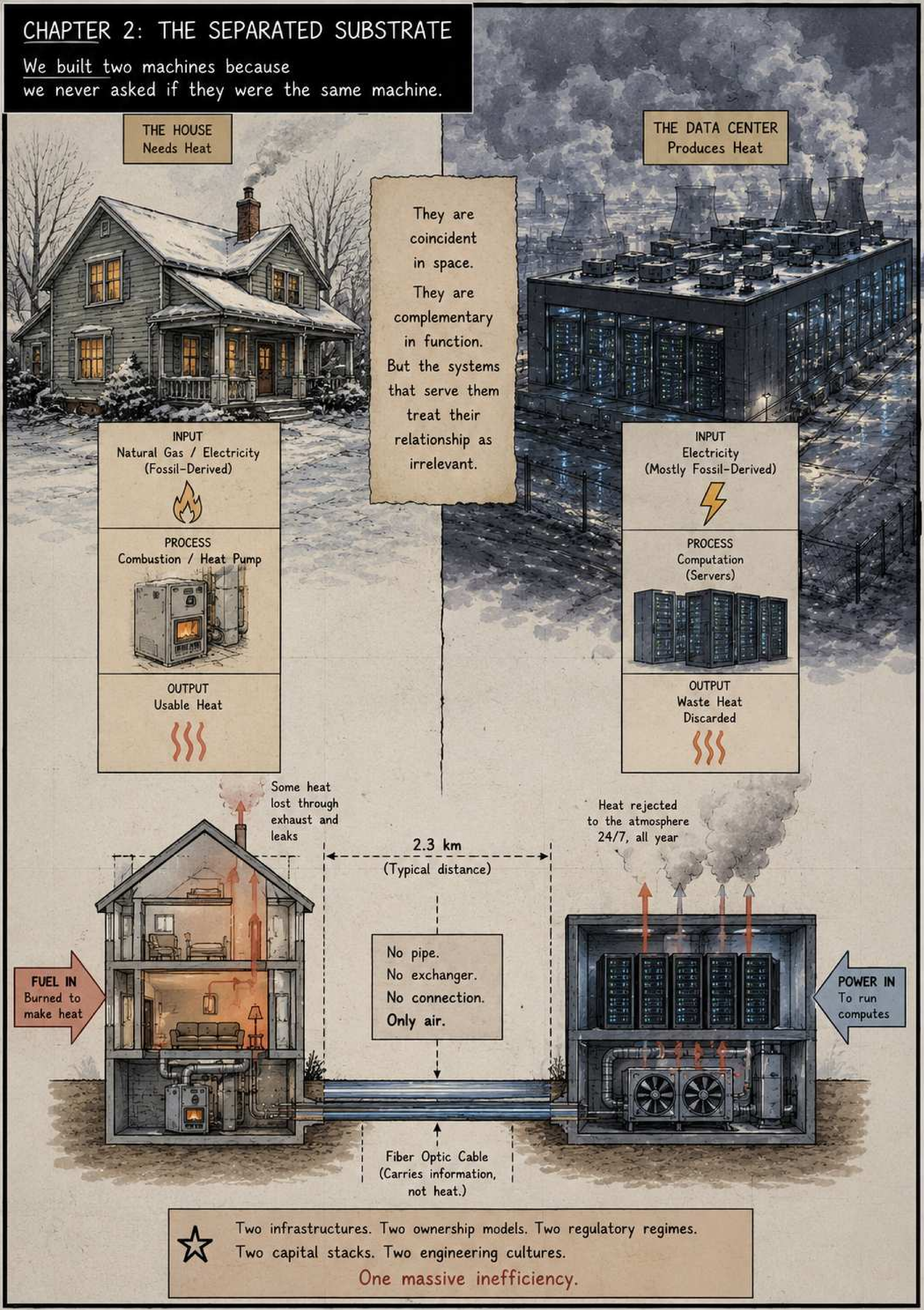
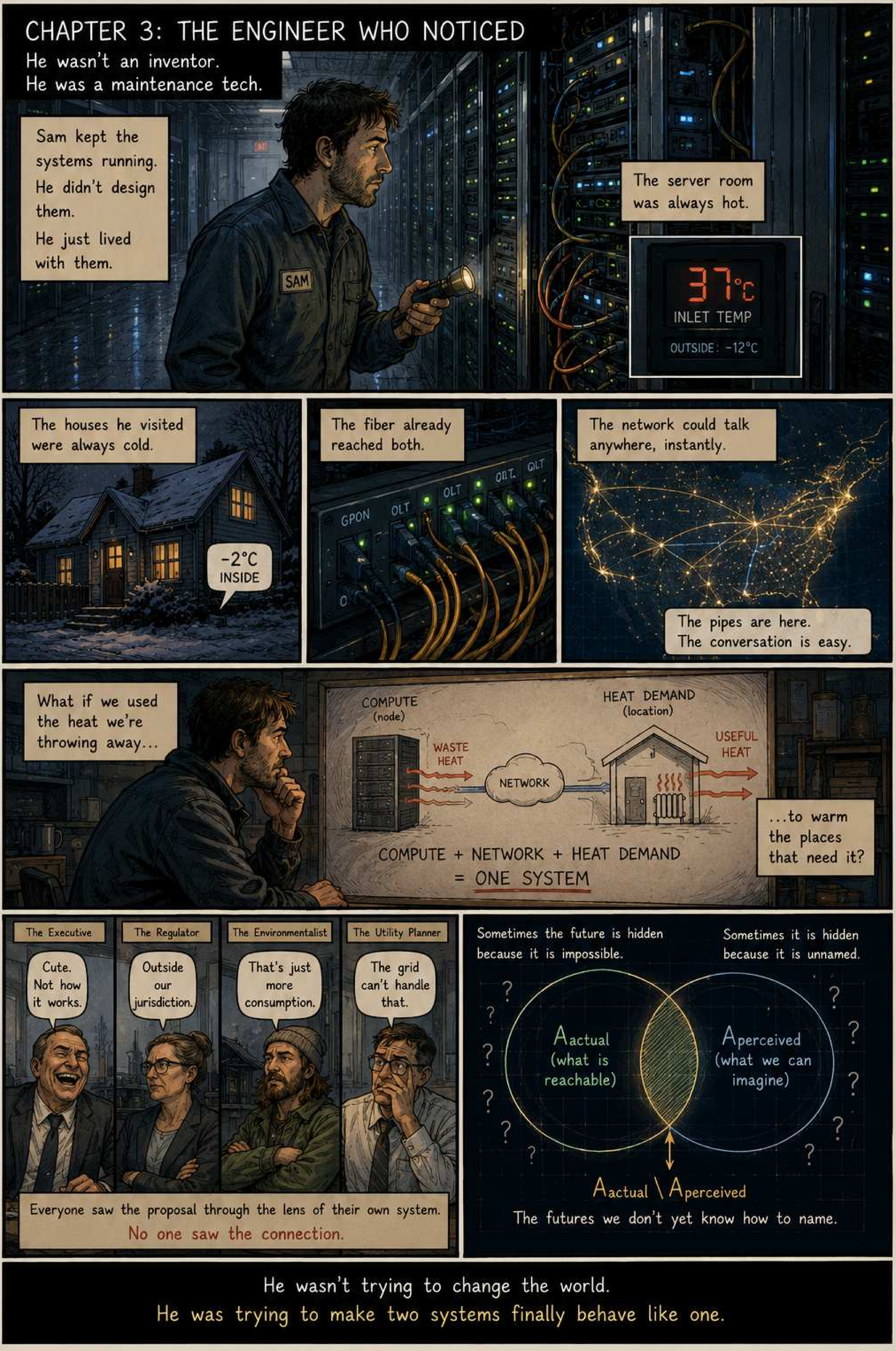
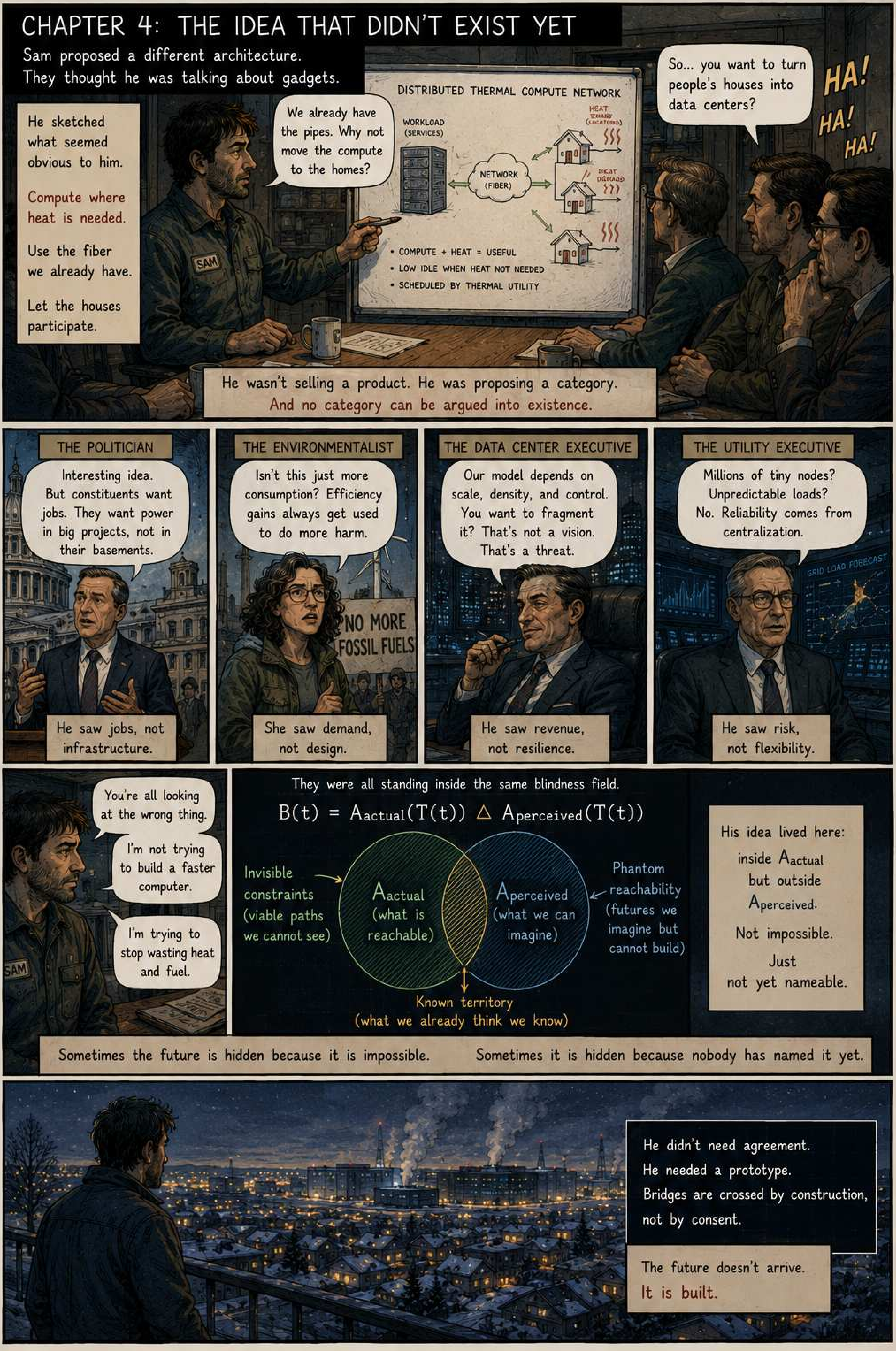
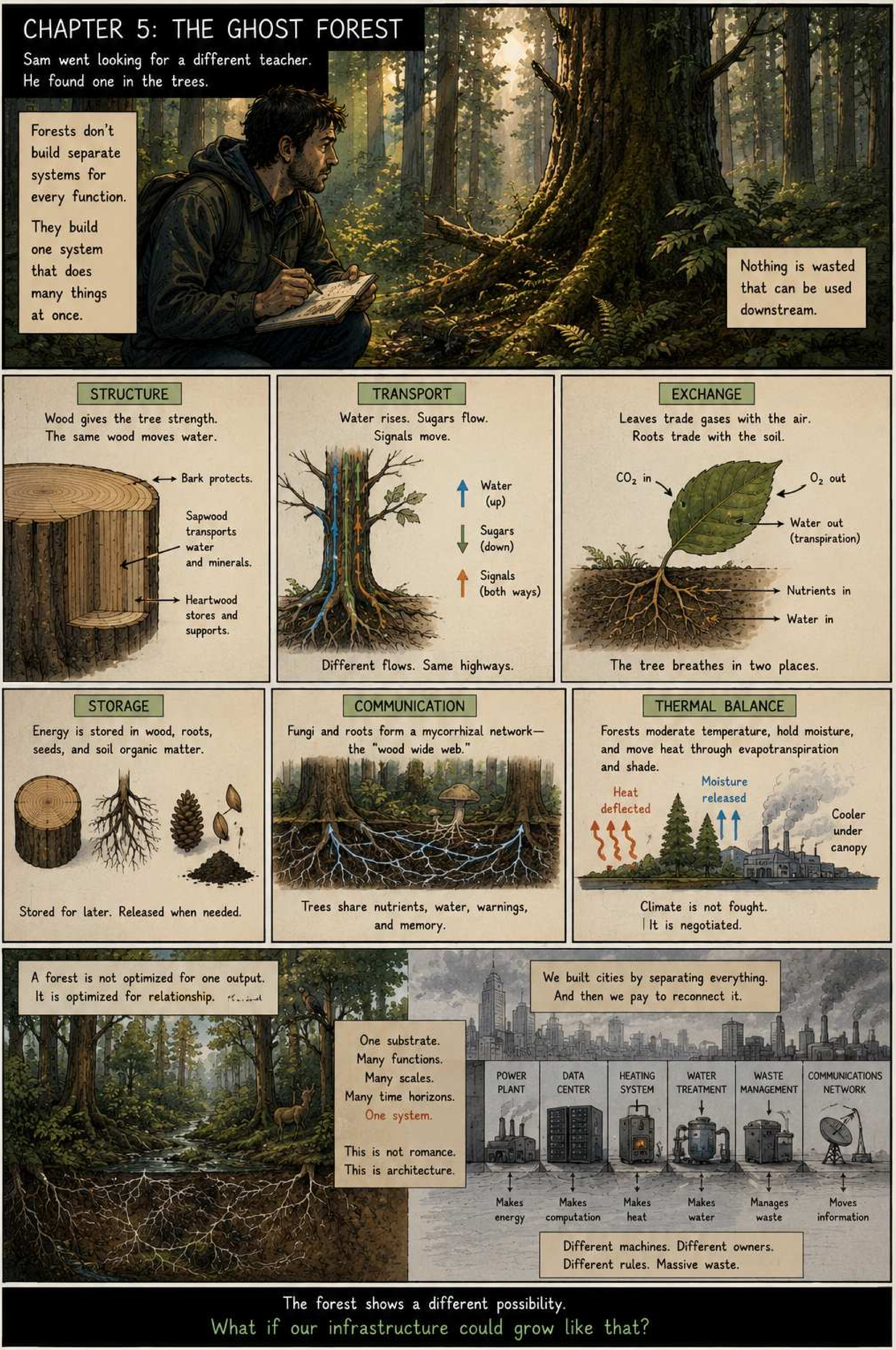

THE HONEST FURNACE
CHAPTER 1: THE FURNACE
Every winter we burn fuel to make heat.
CHUNK
WHOOOOM
22°C
HEATING
Every day we burn electricity to make heat we throw away.
Heat we don't need. Heat we discard.
NORTHSTAR DATA CENTER
FACILITY STATUS
POWER DRAW
48.7 MW
PUE
1.68
HEAT REJECTED
82.3 MW
Two systems. One needs heat. One throws heat away.
No connection.

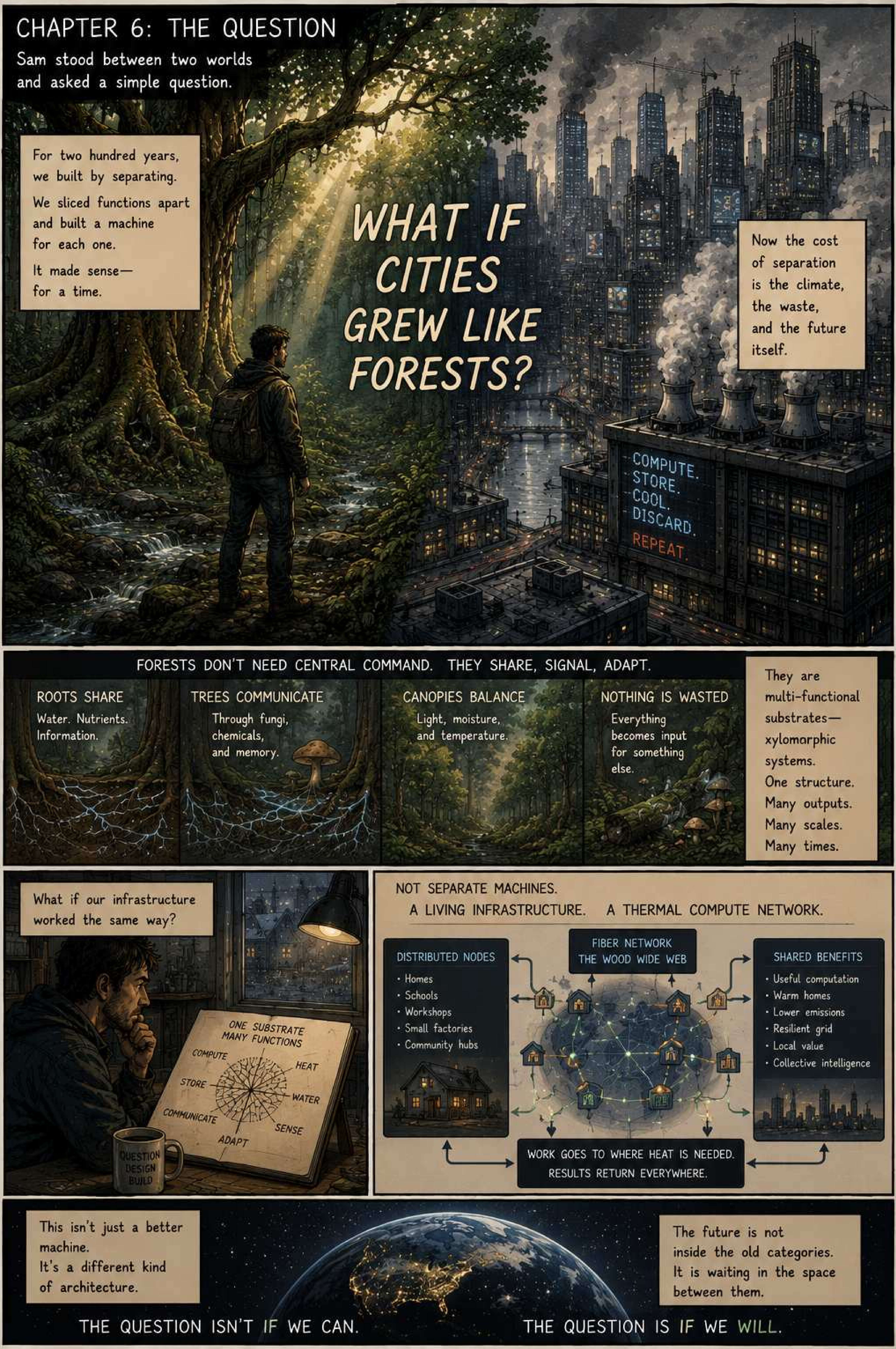

CHAPTER 2: THE SEPARATED SUBSTRATE
We built two machines because
we never asked if they were the same machine.
THE HOUSE
Needs Heat
THE DATA CENTER
Produces Heat
They are coincident in space.
They are complementary in function.
But the systems that serve them treat their relationship as irrelevant.
INPUT
Natural Gas / Electricity
(Fossil-Derived)
PROCESS
Combustion / Heat Pump
OUTPUT
Usable Heat
INPUT
Electricity
(Mostly Fossil-Derived)
PROCESS
Computation
(Servers)
OUTPUT
Waste Heat
Discarded
Some heat lost through exhaust and leaks
2.3 km
(Typical distance)
Heat rejected to the atmosphere 24/7, all year
FUEL IN
Burned to make heat
No pipe.
No exchanger.
No connection.
Only air.
POWER IN
To run computes
Fiber Optic Cable
(Carries information, not heat.)
Two infrastructures. Two ownership models. Two regulatory regimes.
Two capital stacks. Two engineering cultures.
One massive inefficiency.

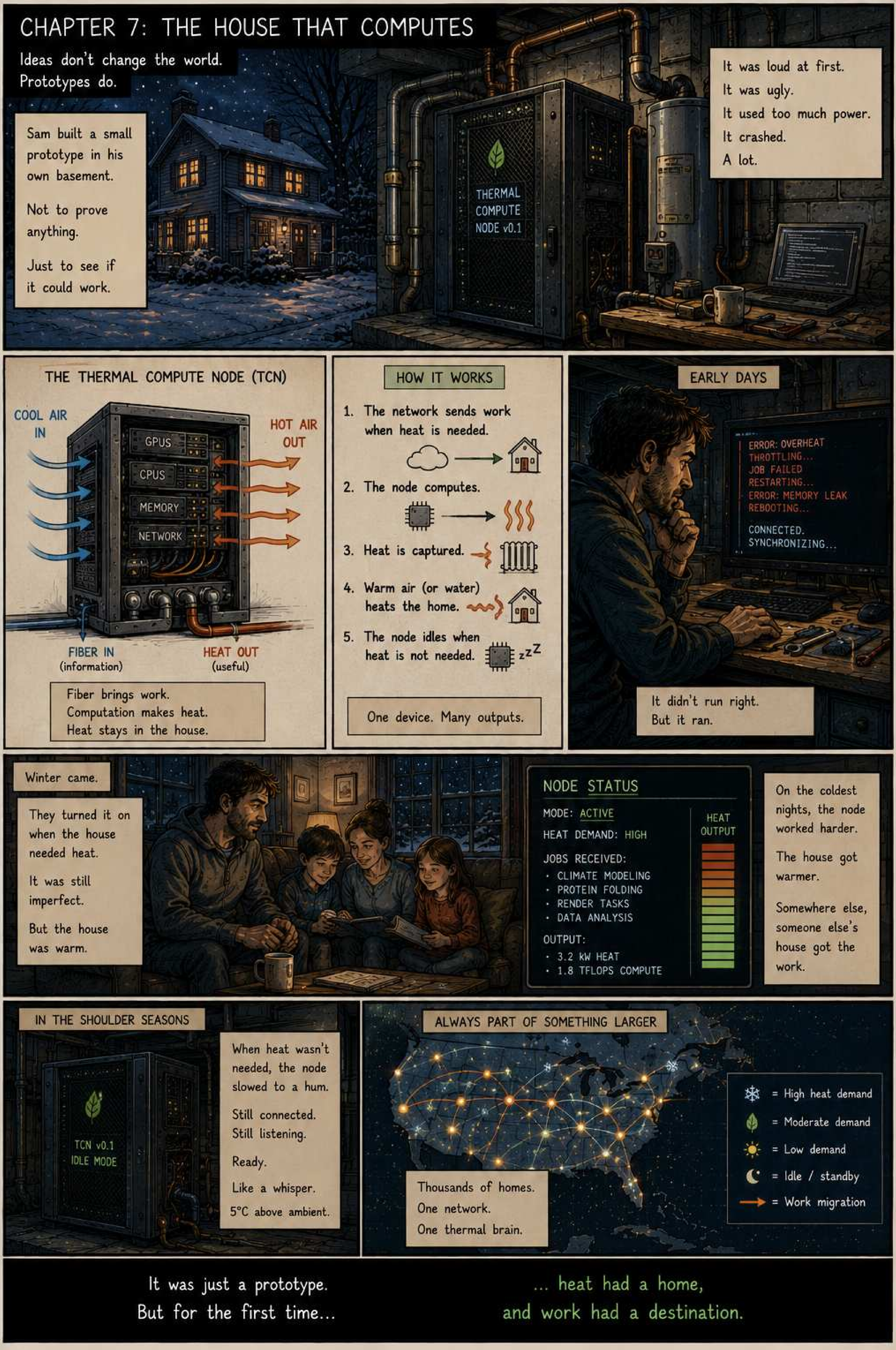

CHAPTER 3: THE ENGINEER WHO NOTICED
He wasn't an inventor.
He was a maintenance tech.
Sam kept the systems running. He didn't design them. He just lived with them.
SAM
The server room was always hot.
37°C
INLET TEMP
OUTSIDE: -12°C
The houses he visited were always cold.
-2°C INSIDE
The fiber already reached both.
GPON
OLT
OLT
OLT
OLT
The network could talk anywhere, instantly.
The pipes are here.
The conversation is easy.
What if we used the heat we're throwing away...
COMPUTE (node)
WASTE HEAT
NETWORK
HEAT DEMAND (location)
USEFUL HEAT
COMPUTE + NETWORK + HEAT DEMAND
= ONE SYSTEM
...to warm the places that need it?
The Executive
Cute. Not how it works.
The Regulator
Outside our jurisdiction.
The Environmentalist
That's just more consumption.
The Utility Planner
The grid can't handle that.
Everyone saw the proposal through the lens of their own system.
No one saw the connection.
Sometimes the future is hidden because it is impossible.
Sometimes it is hidden because it is unnamed.
A_{actual} (what is reachable)
$A_{perceived}$ (what we can imagine)
$A_{actual} \setminus A_{perceived}$
The futures we don't yet know how to name.
He wasn't trying to change the world.
He was trying to make two systems finally behave like one.


CHAPTER 4: THE IDEA THAT DIDN'T EXIST YET
Sam proposed a different architecture.
They thought he was talking about gadgets.
He sketched what seemed obvious to him.
Compute where heat is needed.
Use the fiber we already have.
Let the houses participate.
We already have the pipes. Why not move the compute to the homes?
DISTRIBUTED THERMAL COMPUTE NETWORK
WORKLOAD (SERVICES)
NETWORK (FIBER)
• COMPUTE + HEAT = USEFUL
• LOW IDLE WHEN HEAT NOT NEEDED
• SCHEDULED BY THERMAL UTILITY
SAM
So... you want to turn people's houses into data centers?
HA! HA! HA!
He wasn't selling a product. He was proposing a category.
And no category can be argued into existence.
THE POLITICIAN
Interesting idea. But constituents want jobs. They want power in big projects, not in their basements.
He saw jobs, not infrastructure.
THE ENVIRONMENTALIST
Isn't this just more consumption? Efficiency gains always get used to do more harm.
NO MORE FOSSIL FUELS
She saw demand, not design.
THE DATA CENTER EXECUTIVE
Our model depends on scale, density, and control. You want to fragment it? That's not a vision. That's a threat.
He saw revenue, not resilience.
THE UTILITY EXECUTIVE
Millions of tiny nodes? Unpredictable loads? No. Reliability comes from centralization.
GRID LOAD FORECAST
He saw risk, not flexibility.
You're all looking at the wrong thing.
I'm not trying to build a faster computer.
I'm trying to stop wasting heat and fuel.
SAM
They were all standing inside the same blindness field.
$B(t) = A_{\text{actual}}(T(t)) \triangle A_{\text{perceived}}(T(t))$
Invisible constraints (viable paths we cannot see)
A_{actual} (what is reachable)
$A_{\text{perceived}}$ (what we can imagine)
Phantom reachability (futures we imagine but cannot build)
Known territory (what we already think we know)
His idea lived here: inside A_{actual} but outside $A_{\text{perceived}}$.
Not impossible.
Just not yet nameable.
Sometimes the future is hidden because it is impossible.
Sometimes it is hidden because nobody has named it yet.
He didn't need agreement.
He needed a prototype.
Bridges are crossed by construction, not by consent.
The future doesn't arrive.
It is built.

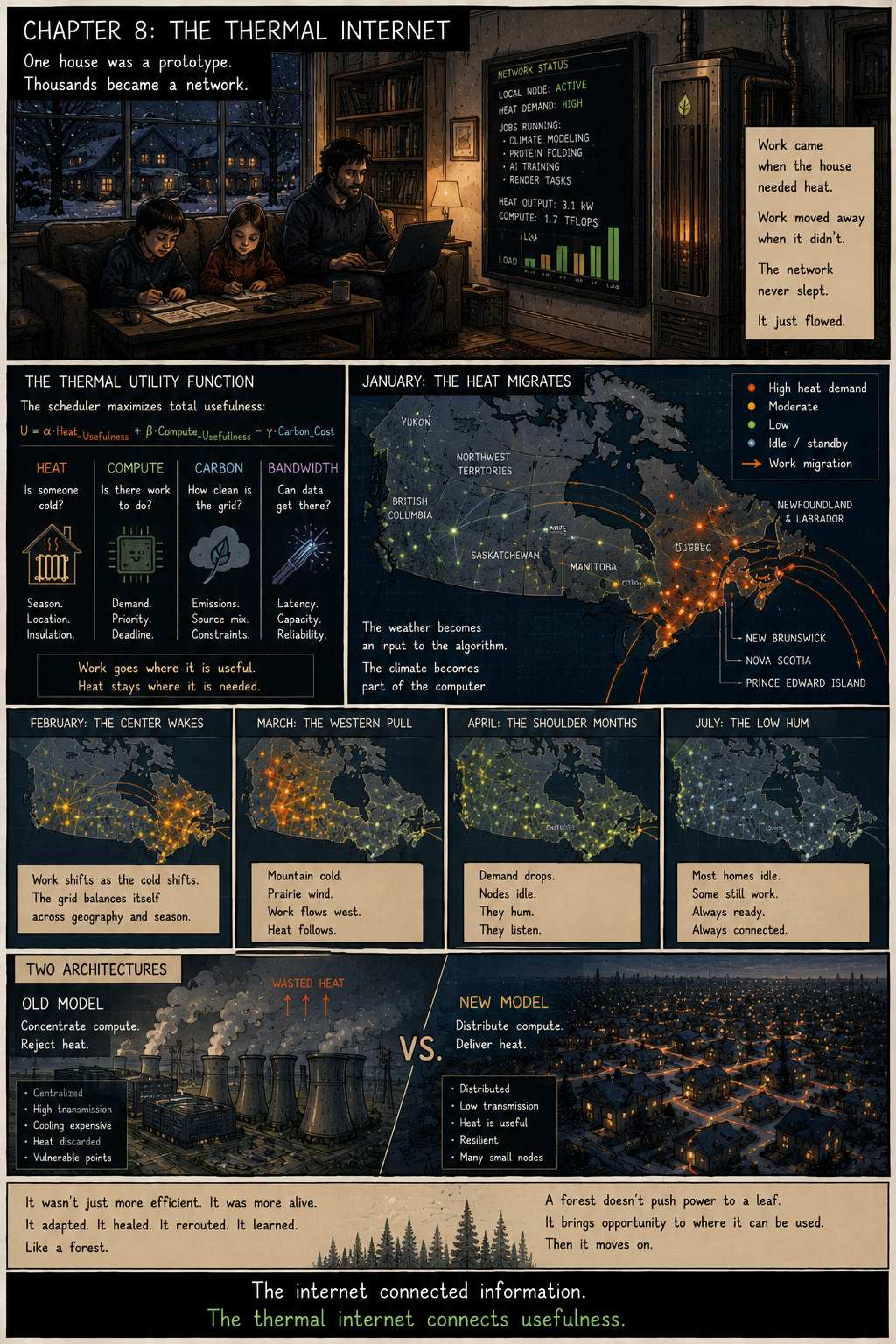


CHAPTER 5: THE GHOST FOREST
Sam went looking for a different teacher. He found one in the trees.
Forests don't build separate systems for every function.
They build one system that does many things at once.
Nothing is wasted that can be used downstream.
STRUCTURE
Wood gives the tree strength. The same wood moves water.
Bark protects.
Sapwood transports water and minerals.
Heartwood stores and supports.
TRANSPORT
Water rises. Sugars flow. Signals move.
Water (up)
Sugars (down)
Signals (both ways)
Different flows. Same highways.
EXCHANGE
Leaves trade gases with the air. Roots trade with the soil.
CO_2 in
O_2 out
Water out (transpiration)
Nutrients in
Water in
The tree breathes in two places.
STORAGE
Energy is stored in wood, roots, seeds, and soil organic matter.
Stored for later. Released when needed.
COMMUNICATION
Fungi and roots form a mycorrhizal network— the "wood wide web."
Trees share nutrients, water, warnings, and memory.
THERMAL BALANCE
Forests moderate temperature, hold moisture, and move heat through evapotranspiration and shade.
Heat deflected
Moisture released
Cooler under canopy
Climate is not fought. It is negotiated.
A forest is not optimized for one output. It is optimized for relationship.
One substrate. Many functions. Many scales. Many time horizons. One system.
This is not romance. This is architecture.
We built cities by separating everything. And then we pay to reconnect it.
POWER PLANT
DATA CENTER
HEATING SYSTEM
WATER TREATMENT
WASTE MANAGEMENT
COMMUNICATIONS NETWORK
Makes energy
Makes computation
Makes heat
Makes water
Manages waste
Moves information
Different machines. Different owners. Different rules. Massive waste.
The forest shows a different possibility.
What if our infrastructure could grow like that?

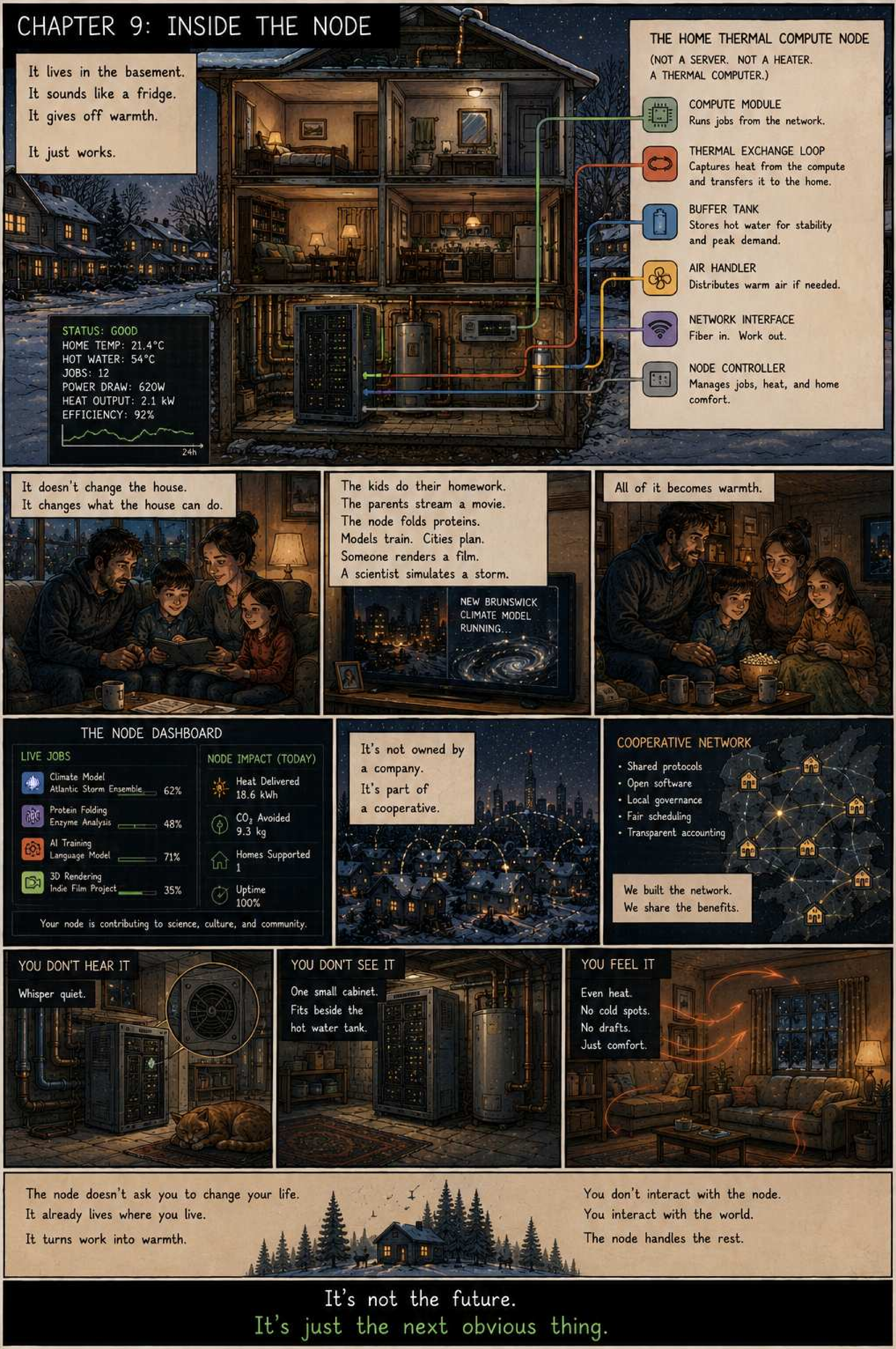

CHAPTER 6: THE QUESTION
Sam stood between two worlds and asked a simple question.
For two hundred years, we built by separating.
We sliced functions apart and built a machine for each one.
It made sense— for a time.
WHAT IF CITIES GREW LIKE FORESTS?
Now the cost of separation is the climate, the waste, and the future itself.
COMPUTE. STORE. COOL. DISCARD. REPEAT.
FORESTS DON'T NEED CENTRAL COMMAND. THEY SHARE, SIGNAL, ADAPT.
ROOTS SHARE
Water. Nutrients. Information.
TREES COMMUNICATE
Through fungi, chemicals, and memory.
CANOPIES BALANCE
Light, moisture, and temperature.
NOTHING IS WASTED
Everything becomes input for something else.
They are multi-functional substrates— xylomorphic systems.
One structure. Many outputs. Many scales. Many times.
What if our infrastructure worked the same way?
ONE SUBSTRATE MANY FUNCTIONS
COMPUTE
HEAT
STORE
WATER
COMMUNICATE
SENSE
ADAPT
QUESTION DESIGN BUILD
NOT SEPARATE MACHINES.
A LIVING INFRASTRUCTURE. A THERMAL COMPUTE NETWORK.
FIBER NETWORK
THE WOOD WIDE WEB
DISTRIBUTED NODES
· Homes
· Schools
· Workshops
· Small factories
· Community hubs
SHARED BENEFITS
· Useful computation
· Warm homes
· Lower emissions
· Resilient grid
· Local value
· Collective intelligence
WORK GOES TO WHERE HEAT IS NEEDED.
RESULTS RETURN EVERYWHERE.
This isn't just a better machine.
It's a different kind of architecture.
The future is not inside the old categories.
It is waiting in the space between them.
THE QUESTION ISN'T IF WE CAN.
THE QUESTION IS IF WE WILL.



CHAPTER 7: THE HOUSE THAT COMPUTES
Ideas don't change the world. Prototypes do.
Sam built a small prototype in his own basement.
Not to prove anything.
Just to see if it could work.
THERMAL COMPUTE NODE v0.1
It was loud at first.
It was ugly.
It used too much power.
It crashed.
A lot.
THE THERMAL COMPUTE NODE (TCN)
COOL AIR IN
HOT AIR OUT
GPUS
CPUS
MEMORY
NETWORK
FIBER IN (information)
HEAT OUT (useful)
Fiber brings work.
Computation makes heat.
Heat stays in the house.
HOW IT WORKS
1. The network sends work when heat is needed.
2. The node computes.
3. Heat is captured.
4. Warm air (or water) heats the home.
5. The node idles when heat is not needed.
zzZ
One device. Many outputs.
EARLY DAYS
ERROR: OVERHEAT
THROTTLING...
JOB FAILED
RESTARTING...
ERROR: MEMORY LEAK
REBOOTING...
CONNECTED.
SYNCHRONIZING...
It didn't run right.
But it ran.
Winter came.
They turned it on when the house needed heat.
It was still imperfect.
But the house was warm.
NODE STATUS
MODE: ACTIVE
HEAT DEMAND: HIGH
JOBS RECEIVED:
• CLIMATE MODELING
• PROTEIN FOLDING
• RENDER TASKS
• DATA ANALYSIS
OUTPUT:
• 3.2 kW HEAT
• 1.8 TFLOPS COMPUTE
HEAT OUTPUT
On the coldest nights, the node worked harder.
The house got warmer.
Somewhere else, someone else's house got the work.
IN THE SHOULDER SEASONS
TCN v0.1 IDLE MODE
When heat wasn't needed, the node slowed to a hum.
Still connected.
Still listening.
Ready.
Like a whisper.
5°C above ambient.
ALWAYS PART OF SOMETHING LARGER
= High heat demand
= Moderate demand
= Low demand
= Idle / standby
= Work migration
Thousands of homes.
One network.
One thermal brain.
It was just a prototype.
But for the first time...
... heat had a home,
and work had a destination.



CHAPTER 8: THE THERMAL INTERNET
One house was a prototype.
Thousands became a network.
NETWORK STATUS
LOCAL NODE: ACTIVE
HEAT DEMAND: HIGH
JOBS RUNNING:
• CLIMATE MODELING
• PROTEIN FOLDING
• AI TRAINING
• RENDER TASKS
HEAT OUTPUT: 3.1 kW
COMPUTE: 1.7 TFLOPS
LOAD
Work came when the house needed heat.
Work moved away when it didn't.
The network never slept.
It just flowed.
THE THERMAL UTILITY FUNCTION
The scheduler maximizes total usefulness:
$U = \alpha \cdot \text{Heat}_{\text{Usefulness}} + \beta \cdot \text{Compute}_{\text{Usefulness}} - \gamma \cdot \text{Carbon\_Cost}$
HEAT
Is someone cold?
Season. Location. Insulation.
COMPUTE
Is there work to do?
Demand. Priority. Deadline.
CARBON
How clean is the grid?
Emissions. Source mix. Constraints.
BANDWIDTH
Can data get there?
Latency. Capacity. Reliability.
Work goes where it is useful.
Heat stays where it is needed.
JANUARY: THE HEAT MIGRATES
High heat demand
Moderate
Low
Idle / standby
Work migration
YUKON
NORTHWEST TERRITORIES
BRITISH COLUMBIA
SASKATCHEWAN
MANITOBA
QUEBEC
NEWFOUNDLAND & LABRADOR
NEW BRUNSWICK
NOVA SCOTIA
PRINCE EDWARD ISLAND
The weather becomes an input to the algorithm.
The climate becomes part of the computer.
FEBRUARY: THE CENTER WAKES
Work shifts as the cold shifts.
The grid balances itself across geography and season.
MARCH: THE WESTERN PULL
Mountain cold.
Prairie wind.
Work flows west.
Heat follows.
APRIL: THE SHOULDER MONTHS
Demand drops.
Nodes idle.
They hum.
They listen.
JULY: THE LOW HUM
Most homes idle.
Some still work.
Always ready.
Always connected.
TWO ARCHITECTURES
OLD MODEL
Concentrate compute.
Reject heat.
WASTED HEAT
• Centralized
• High transmission
• Cooling expensive
• Heat discarded
• Vulnerable points
VS.
NEW MODEL
Distribute compute.
Deliver heat.
• Distributed
• Low transmission
• Heat is useful
• Resilient
• Many small nodes
It wasn't just more efficient. It was more alive.
It adapted. It healed. It rerouted. It learned.
Like a forest.
A forest doesn't push power to a leaf.
It brings opportunity to where it can be used.
Then it moves on.
The internet connected information.
The thermal internet connects usefulness.



CHAPTER 9: INSIDE THE NODE
It lives in the basement.
It sounds like a fridge.
It gives off warmth.
It just works.
THE HOME THERMAL COMPUTE NODE
(NOT A SERVER. NOT A HEATER. A THERMAL COMPUTER.)
COMPUTE MODULE
Runs jobs from the network.
THERMAL EXCHANGE LOOP
Captures heat from the compute and transfers it to the home.
BUFFER TANK
Stores hot water for stability and peak demand.
AIR HANDLER
Distributes warm air if needed.
NETWORK INTERFACE
Fiber in. Work out.
NODE CONTROLLER
Manages jobs, heat, and home comfort.
STATUS: GOOD
HOME TEMP: 21.4°C
HOT WATER: 54°C
JOBS: 12
POWER DRAW: 620W
HEAT OUTPUT: 2.1 kW
EFFICIENCY: 92%
24h
It doesn't change the house.
It changes what the house can do.
The kids do their homework.
The parents stream a movie.
The node folds proteins.
Models train. Cities plan.
Someone renders a film.
A scientist simulates a storm.
NEW BRUNSWICK CLIMATE MODEL RUNNING...
All of it becomes warmth.
THE NODE DASHBOARD
LIVE JOBS
Climate Model Atlantic Storm Ensemble 62%
Protein Folding Enzyme Analysis 48%
AI Training Language Model 71%
3D Rendering Indie Film Project 35%
NODE IMPACT (TODAY)
Heat Delivered 18.6 kWh
CO₂ Avoided 9.3 kg
Homes Supported 1
Uptime 100%
Your node is contributing to science, culture, and community.
It's not owned by a company.
It's part of a cooperative.
COOPERATIVE NETWORK
• Shared protocols
• Open software
• Local governance
• Fair scheduling
• Transparent accounting
We built the network.
We share the benefits.
YOU DON'T HEAR IT
Whisper quiet.
YOU DON'T SEE IT
One small cabinet.
Fits beside the hot water tank.
YOU FEEL IT
Even heat.
No cold spots.
No drafts.
Just comfort.
The node doesn't ask you to change your life.
It already lives where you live.
It turns work into warmth.
You don't interact with the node.
You interact with the world.
The node handles the rest.
It's not the future.
It's just the next obvious thing.

THE NEXT OBVIOUS THING

DISTRIBUTED. INTELLIGENT. ALIVE.

Hundreds of small compute-thermal spheres work together to keep your home warm, your systems running, and your world mapped. They are heaters, computers, sensors, batteries, and neighbors—all in one.

EACH SPHERE IS:

- Compute
- Memory
- Sensors & LIDAR
- Battery Buffer
- Mesh Network
- Thermal Mass
- Secure Enclave
- Self-Healing

HOW IT WORKS

Sense
Continuously map the home with LIDAR and sensors.

Understand
Build a live 3D model. Detect occupancy, heat loss, airflow, and usage patterns.

Decide
Distribute compute jobs and heat where they are needed most.

Act
Move work. Move heat. Balance comfort, efficiency, and longevity.

Learn
Improve the model. Adapt to seasons, people, and time.

EMBEDDED
Some spheres live in walls, ceilings, and floors.

MOBILE
Spheres move to where they're needed most.

STRATEGIC
Positions near windows, vents, plumbing, and appliances.

HIDDEN
They live in furniture, behind art, under stairs, and in plain sight.

DOCK & CHARGE
Spheres return to docks to charge, share heat, and exchange data.

ALWAYS MAPPING

The home is never static.
Neither is the map.

- Occupancy
- Movement
- Heat leaks
- Airflow
- Changes over time

All updated. All local.
All the time.

THE THERMAL COLUMN

WINTER	SPRING / FALL	SUMMER
Heat demand high. Column charges and glows.	Balanced. Column breathes. Work shifts.	Heat demand low. Column rests. Mostly idle.

A thermal battery. A compute spine. A living chimney.

RESILIENT BY NATURE

No single point of failure.
No central server.
The network heals, reroutes, and continues.

DAY IN THE LIFE

- 6:30 Wake. Warm the rooms you use.
- 9:00 Work. Compute shifts where you are.
- 12:00 Cook. Ventilate. Balance.
- 18:00 Relax. Comfort follows your presence.
- 23:00 Sleep. System settles into quiet efficiency.

BUILT FOR WHAT MATTERS

- Comfort — without waste
- Privacy — by design
- Efficiency — that learns
- Longevity — that adapts
- Sustainability — that compounds

BY PEOPLE, FOR PEOPLE

Designed. Built. Operated.
Not by a company.
By a cooperative.
For the homes.
For the future.

NOT JUST A HEATER.
NOT JUST A COMPUTER.

AN INFRASTRUCTURE THAT CARES.

Small spheres. Vast network.
Shared purpose.
A home that works with you.

LEGEND

Heat flow · Compute flow · Data / Mesh · Dock / Charge · Idle Sleep · Active Work · Moving

This is not science fiction.
It is the next obvious thing.
We just haven't built it yet.

THERMAL COMPUTE SPHERES

The successor to the furnace. The successor to the data center.
A new kind of object for a new kind of infrastructure.

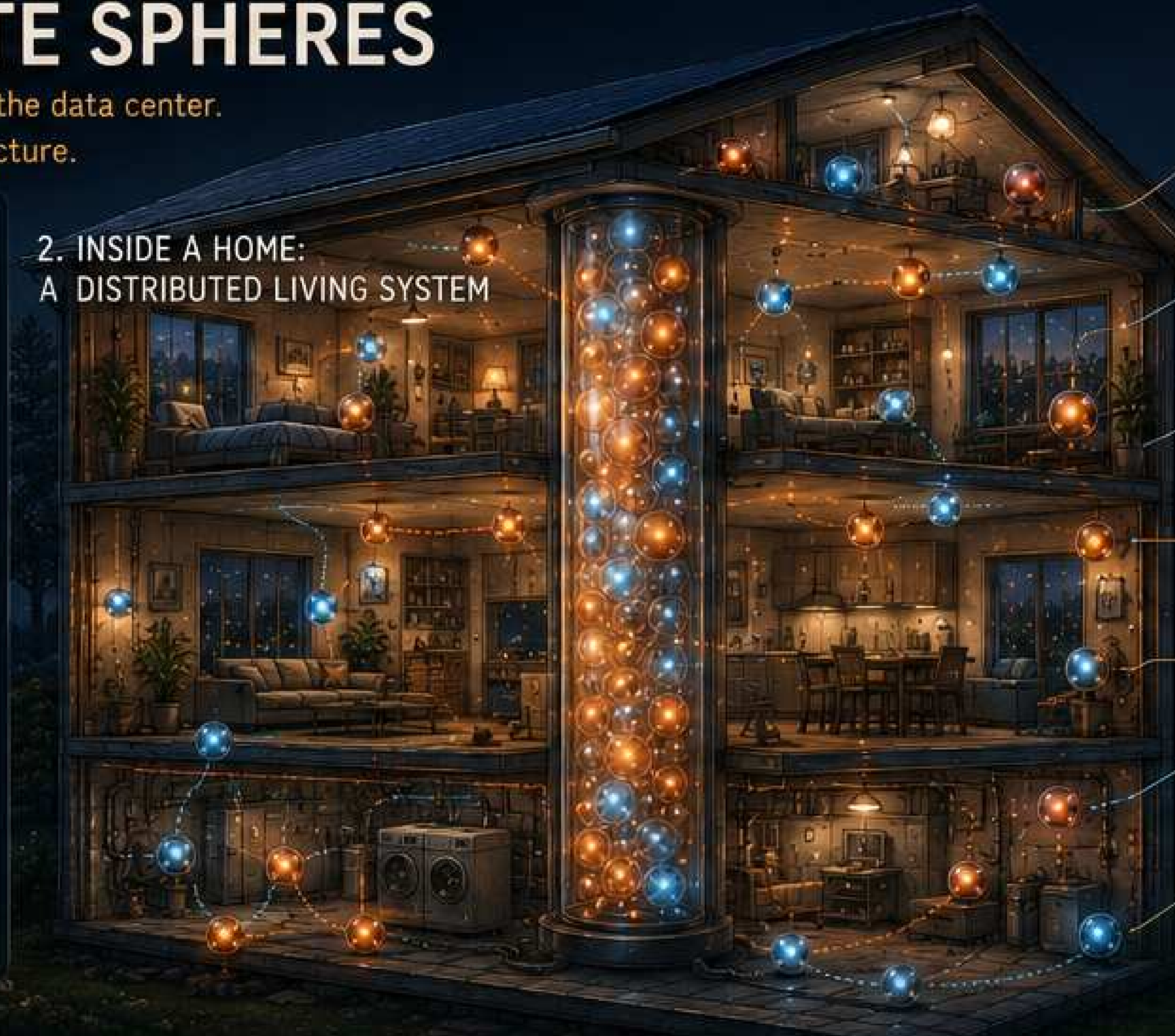
1. WHAT IS A SPHERE?

A complete unit of computation, thermal management, sensing, and connectivity.

- **Compute + Memory** Local processing and storage
- **Thermal Core** Stores, generates, and moves heat
- **Lidar + Sensors** 360° awareness of space and movement
- **Battery Buffer** Ride through outages, balance the grid
- **Mesh Radio** Always connected
- **Dock / I/O** Power, data, plumbing interface

One sphere. Many roles.
Working with the others.

2. INSIDE A HOME: A DISTRIBUTED LIVING SYSTEM



Active (computing / heating) · Standby / buffer · Dormant / sleeping · Mesh network (data + heat)

3. ALWAYS MAPPING

Every sphere sees. The home maintains a real-time 3D model with lidar.

Occupancy · Movement · Heat leaks · Airflow · Changes over time

The map improves. The system learns.
The house becomes self-aware
(geometric awareness, not consciousness).

4. THE SPHERES WORK TOGETHER

Jobs, heat, and data flow across the mesh.
The network balances itself.

- Load balancing
- Heat sharing
- Data routing
- Fault tolerance
- Self-healing

No single point of failure.
The whole is the system.

5. THE THERMAL COLUMN

A visible thermal battery and compute spine.

- **WINTER MODE** Store heat. Do more work. Keep the home warm.
- **SHOULDER SEASON** Balanced. Adapt to needs.
- **SUMMER MODE** Minimal heat. Shift work elsewhere.
- **STANDBY** Rest. Monitor. Be ready.

The column breathes with the seasons.

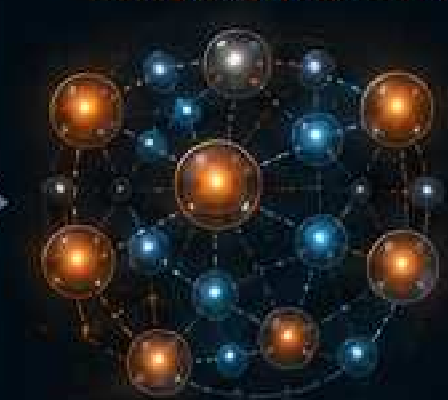
6. FROM MANY DEVICES TO ONE LIVING INFRASTRUCTURE

TODAY (SEPARATE DEVICES)

Furnace · Thermostat (72°) · Router · Security Camera · Smoke Detector · Motion Sensors · Smart Speaker · Battery Backup

Multiple devices. Multiple apps. Multiple failures.

TOMORROW (ONE LIVING SYSTEM)



- Heating
- Cooling
- Compute
- Storage
- Sensing (lidar)
- Connectivity
- Power buffering
- Health monitoring
- Security
- Adaptation

One system. Many abilities. Infinite configurations.

7. PLANETARY SCALE

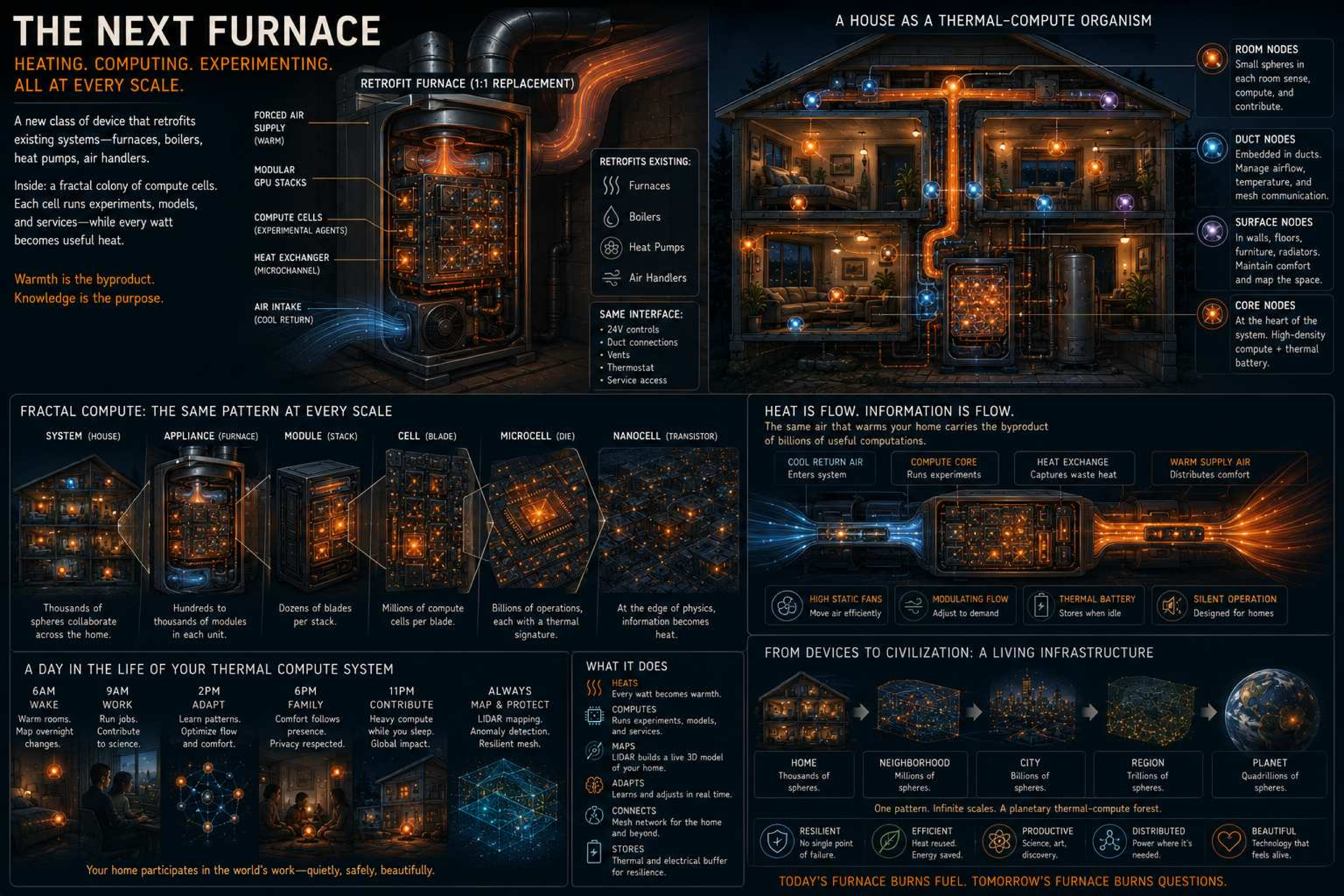
Spheres don't just live in homes.
They form a planetary thermal-compute forest.

Excess heat becomes useful work · Work migrates with climate · Resilient, local, and cooperative · A global system that cares

PRINCIPLES THAT GUIDE THE DESIGN

- **LOCAL IS RESILIENT** Every place can stand on its own.
- **SHARE WHAT HELPS** Heat, compute, and data are shared resources.
- **ADAPT CONTINUOUSLY** The system learns and evolves.
- **BE AUTONOMOUS, NOT ISOLATED** Cooperate. Don't depend.
- **CARE FOR THE WHOLE** The goal is a livable planet.

These spheres are small.
But together, they carry what matters most.
We don't just build infrastructure.
We build the future.


THE NEXT FURNACE
HEATING. COMPUTING. EXPERIMENTING.
ALL AT EVERY SCALE.
A new class of device that retrofits existing systems—furnaces, boilers, heat pumps, air handlers.
Inside: a fractal colony of compute cells. Each cell runs experiments, models, and services—while every watt becomes useful heat.
Warmth is the byproduct.
Knowledge is the purpose.
RETROFIT FURNACE (1:1 REPLACEMENT)
FORCED AIR SUPPLY (WARM)
MODULAR GPU STACKS
COMPUTE CELLS (EXPERIMENTAL AGENTS)
HEAT EXCHANGER (MICROCHANNEL)
AIR INTAKE (COOL RETURN)
RETROFITS EXISTING:
Furnaces
Boilers
Heat Pumps
Air Handlers
SAME INTERFACE:
• 24V controls
• Duct connections
• Vents
• Thermostat
• Service access
A HOUSE AS A THERMAL-COMPUTE ORGANISM
ROOM NODES
Small spheres in each room sense, compute, and contribute.
DUCT NODES
Embedded in ducts. Manage airflow, temperature, and mesh communication.
SURFACE NODES
In walls, floors, furniture, radiators. Maintain comfort and map the space.
CORE NODES
At the heart of the system. High-density compute + thermal battery.
FRACTAL COMPUTE: THE SAME PATTERN AT EVERY SCALE
SYSTEM (HOUSE)
Thousands of spheres collaborate across the home.
APPLIANCE (FURNACE)
Hundreds to thousands of modules in each unit.
MODULE (STACK)
Dozens of blades per stack.
CELL (BLADE)
Millions of compute cells per blade.
MICROCELL (DIE)
Billions of operations, each with a thermal signature.
NANOCELL (TRANSISTOR)
At the edge of physics, information becomes heat.
HEAT IS FLOW. INFORMATION IS FLOW.
The same air that warms your home carries the byproduct of billions of useful computations.
COOL RETURN AIR
Enters system
COMPUTE CORE
Runs experiments
HEAT EXCHANGE
Captures waste heat
WARM SUPPLY AIR
Distributes comfort
HIGH STATIC FANS
Move air efficiently
MODULATING FLOW
Adjust to demand
THERMAL BATTERY
Stores when idle
SILENT OPERATION
Designed for homes
A DAY IN THE LIFE OF YOUR THERMAL COMPUTE SYSTEM
6AM
WAKE
Warm rooms. Map overnight changes.
9AM
WORK
Run jobs. Contribute to science.
2PM
ADAPT
Learn patterns. Optimize flow and comfort.
6PM
FAMILY
Comfort follows presence. Privacy respected.
11PM
CONTRIBUTE
Heavy compute while you sleep. Global impact.
ALWAYS
MAP & PROTECT
LIDAR mapping. Anomaly detection. Resilient mesh.
Your home participates in the world's work—quietly, safely, beautifully.
WHAT IT DOES
HEATS
Every watt becomes warmth.
COMPUTES
Runs experiments, models, and services.
MAPS
LIDAR builds a live 3D model of your home.
ADAPTS
Learns and adjusts in real time.
CONNECTS
Mesh network for the home and beyond.
STORES
Thermal and electrical buffer for resilience.
FROM DEVICES TO CIVILIZATION: A LIVING INFRASTRUCTURE
HOME
Thousands of spheres.
NEIGHBORHOOD
Millions of spheres.
CITY
Billions of spheres.
REGION
Trillions of spheres.
PLANET
Quadrillions of spheres.
One pattern. Infinite scales. A planetary thermal-compute forest.
RESILIENT
No single point of failure.
EFFICIENT
Heat reused. Energy saved.
PRODUCTIVE
Science, art, discovery.
DISTRIBUTED
Power where it's needed.
BEAUTIFUL
Technology that feels alive.
TODAY'S FURNACE BURNS FUEL. TOMORROW'S FURNACE BURNS QUESTIONS.

Xylomorphic Infrastructure: Compute, Heat, and Resources in Every Cell

A civilization built from recursive local agents that exchange matter, energy, and information.

1. THERMAL-COMPUTE FURNACE (RETROFIT)

Looks like today's furnace.
Inside: a fractal colony of compute cells.

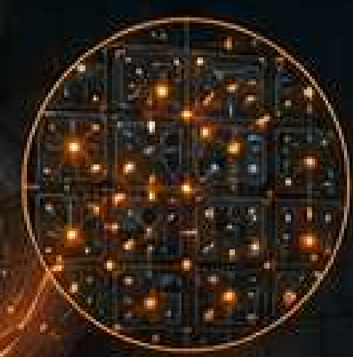
FORCED AIR OUT Warm air to home

COMPUTE COLONY Thousands of compute cells running experiments, training, and services

HEAT EXCHANGER Captures waste heat from computation

AIR INTAKE Cool return air from home

FRACTAL AT EVERY SCALE
Cell → Module → Stack → Unit



EVERY CELL CAN:
- Run experiments
- Train models
- Store and route data
- Sense and map
- Collaborate
- Learn and adapt

Every watt becomes heat.
Every joule becomes knowledge.

2. THE HOME AS A LIVING INFRASTRUCTURE

Hundreds of spheres. One cooperative organism.

ROOM SPHERES Comfort, sensing, mapping

DUCT SPHERES Airflow, temperature, humidity

SURFACE SPHERES In walls, furniture, radiators

CORE SPHERES High-density compute + thermal battery

UTILITY SPHERES Water, electrical, plumbing systems

MOBILE SPHERES Move to where needed

WHAT THEY DO

Heat the home · Compute and learn · Map in 3D (lidar) · Route data · Balance the mesh · Store energy · Adapt in real time

3. ALWAYS MAPPING

A live 3D model of the home and its systems.
Every sphere sees a little. Together they see everything.

- Occupancy
- Movement
- Heat leaks
- Airflow
- Changes over time
- Anomaly detection

4. SPHERES WORK TOGETHER (STIGMERGY)

No central controller. Local signals. Global order.

- Share work
- Share heat
- Share data
- Self-organize
- Self-heal

5. LOCAL RESOURCE CYCLES: GARBAGE COLLECTION AS A STIGMERGIC SYSTEM

Waste is not trash. It is feedstock in the wrong place.

1. **COLLECT** People gather litter and household waste.
2. **PACK** Pack into small resource balls.
3. **PICK UP** Autonomous collectors pick up full balls.
4. **SORT & PROCESS** At local depots, balls are opened, scanned, sorted, and re-packed.
5. **REUSE & MANUFACTURE** Materials flow to local makers and industry.

RESOURCE BALL

Standardized container.

ID: 7A9F-3C2B
Type: Mixed Plastics
Weight: 4.3 kg
Origin: Zone 12-B
Collected: 2025-05-18
Est. Value: $0.72
Route: Depot 7
Status: Sorting

VORONOI TERRITORIES

The city self-organizes into natural catchment cells around local depots.

Depot
Border

You belong to the nearest depot.

6. RECURSIVE AT EVERY SCALE

Same pattern. Different scale.

Cell (microscopic) → Compute Module (centimeters) → Furnace / Unit (meters) → Home (tens of meters) → Neighborhood (hundreds of meters) → District Hub (kilometers) → Region Network (tens of kilometers) → Planetary Mesh (global)

Different materials. Same geometry. Local intelligence. Global cooperation.

7. PARTICIPATE. CONTRIBUTE. EARN.

You get paid to help your system thrive.

Bring resource balls to depots → Provide useful materials → Share storage, compute, and heat → Improve your local network → Earn credits and benefits

Reputation + contribution = access + rewards.
The system grows stronger when everyone participates.

8. THE RESULT

- Less waste.
- Lower emissions.
- Local resilience.
- Beautiful, adaptive cities.
- A civilization that thinks, cleans, builds, and cares —together.

PRINCIPLES THAT GUIDE THE DESIGN

- **LOCAL FIRST** Work happens closest to where it matters.
- **RECURSIVE DESIGN** The same pattern at every scale.
- **STIGMERGY** No boss. Just local signals and shared state.
- **CLOSED LOOPS** Resources circulate. Nothing is waste.
- **BEAUTY & BELONGING** Infrastructure should feel alive.
- **ADAPT & EVOLVE** Systems learn, adapt, and get better over time.
- **CARE FOR THE WHOLE** People, places, and planet together or nothing.

The successor to the furnace was not a better furnace.
The successor to the data center was not a better data center.
Both disappeared into a new kind of object.
A new kind of civilization.